

WEEK-2

Roll number: 2520090041

Name: Pavani

The Librarian's Dashboard Scenario The library is now digital! The head librarian needs a report to see which members are active and which books are in the collection. You also need to handle a "Book Donation" process where a new book is added and assigned to a category at the exact same time—if one part fails, the whole process must stop to keep the catalog accurate.

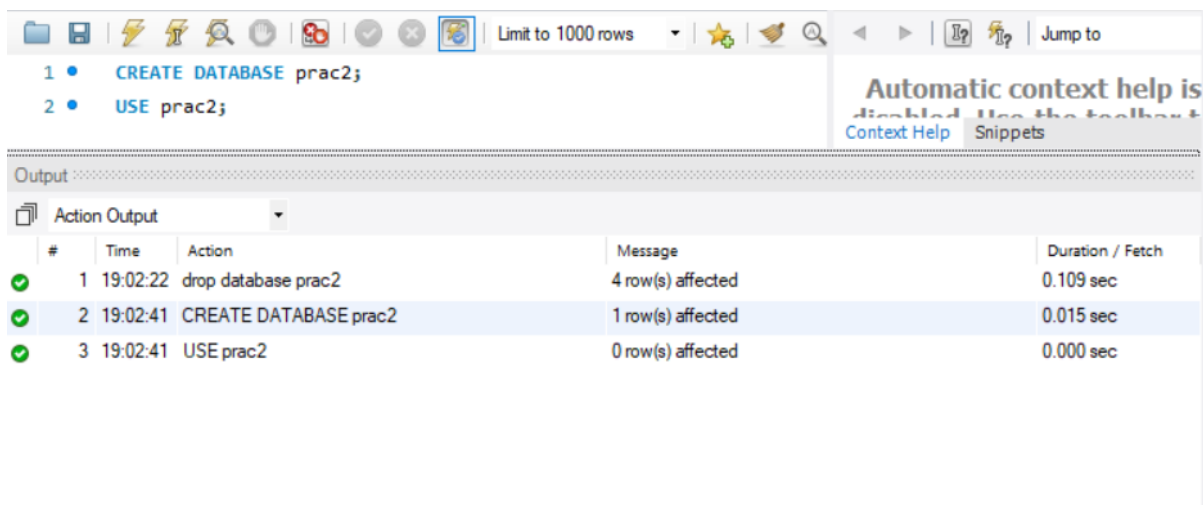
Task to be Done

- 1. The Catalog Search (JOINS):** Create a third table called Loans that connects member_id to book_id. Write a query to show the Member Name and the Book Title they have currently borrowed.
- 2. Collection Stats (Aggregations):** Write a query to find the total number of books published in each year. o Hint: Use COUNT(book_id) and GROUP BY published_year.
- 3. The Donation Transaction (ACID):** Use a Transaction (BEGIN...COMMIT) to add a new book to the Books table and simultaneously log the entry in a Donation_History table.
- 4. Search Speed (Indexing):** Create an Index on the isbn column. Explain why searching by ISBN is now faster for the librarian than searching by Title.

Skill Set Gained

- Relational Logic: Connecting independent tables (Books + Members) through a Junction table (Loans).
- Analytical SQL: Using GROUP BY to generate business insights.
- Atomic Operations: Using Transactions to ensure multiple database changes succeed or fail as a single unit.

Step 1 — Create Database



The screenshot shows a database management interface with a toolbar at the top. Below the toolbar, two SQL commands are entered in a text area:

```
1 • CREATE DATABASE prac2;  
2 • USE prac2;
```

To the right of the text area, there is a notification: "Automatic context help is disabled. Use the toolbar Context Help Snippets".

Below the text area, the "Output" tab is selected, displaying a table of execution results:

#	Time	Action	Message	Duration / Fetch
✓ 1	19:02:22	drop database prac2	4 row(s) affected	0.109 sec
✓ 2	19:02:41	CREATE DATABASE prac2	1 row(s) affected	0.015 sec
✓ 3	19:02:41	USE prac2	0 row(s) affected	0.000 sec

Step 2 — Create Books Table (DDL + Constraints)

The screenshot shows a database IDE interface. The top toolbar includes icons for file operations, execution, and help. A dropdown menu is set to "Limit to 1000 rows". The main editor displays the following SQL code:

```
1 CREATE TABLE Books (  
2     book_id INT PRIMARY KEY,  
3     title VARCHAR(100) NOT NULL,  
4     isbn VARCHAR(20) UNIQUE,  
5     published_year INT CHECK (published_year < 2027)  
6 );
```

On the right, a context help panel is visible with the text: "Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help." Below this are tabs for "Context Help" and "Snippets".

The bottom section is titled "Output" and contains a table of execution results:

#	Time	Action	Message	Duration / Fetch
✓ 1	19:02:22	drop database prac2	4 row(s) affected	0.109 sec
✓ 2	19:02:41	CREATE DATABASE prac2	1 row(s) affected	0.015 sec
✓ 3	19:02:41	USE prac2	0 row(s) affected	0.000 sec
✓ 4	19:04:02	CREATE TABLE Books (book_id INT PRIMARY K...	0 row(s) affected	0.078 sec

Step 3 & 4 — Insert Books and Display

SCHEMAS

Filter objects

- bookflow_db
- bookflow_dbms
- college
- college_db
- college_dbms
- db
- dbms
- sys

Limit to 1000 rows

Jump to

```

1 • INSERT INTO Books (book_id, title, isbn, published_year) VALUES
2   (1, 'The Great Gatsby', '9780743273565', 1925),
3   (2, 'To Kill a Mockingbird', '9780061120084', 1960),
4   (3, '1984', '9780451524935', 1949);
5

```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 1	19:06:49	drop database prac2	1 row(s) affected	0.032 sec
✓ 2	19:07:46	CREATE DATABASE prac2	1 row(s) affected	0.000 sec
✓ 3	19:07:46	USE prac2	0 row(s) affected	0.000 sec
✓ 4	19:07:51	CREATE TABLE Books (book_id INT PRIMARY K...	0 row(s) affected	0.032 sec
✓ 5	19:08:08	INSERT INTO Books (book_id, title, isbn, published_...	3 row(s) affected Records: 3 Duplicates: 0 Warning...	0.015 sec

SCHEMAS

Filter objects

- bookflow_db
- bookflow_dbms
- college
- college_db
- college_dbms
- db
- dbms
- sys

Limit to 1000 rows

Jump to

```

1 • SELECT * FROM Books;
2

```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Result Grid

	book_id	title	isbn	published_year
1	1	The Great Gatsby	9780743273565	1925
2	2	To Kill a Mockingbird	9780061120084	1960
3	3	1984	9780451524935	1949
*	HULL	HULL	HULL	HULL

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 1	19:06:49	drop database prac2	1 row(s) affected	0.032 sec
✓ 2	19:07:46	CREATE DATABASE prac2	1 row(s) affected	0.000 sec
✓ 3	19:07:46	USE prac2	0 row(s) affected	0.000 sec
✓ 4	19:07:51	CREATE TABLE Books (book_id INT PRIMARY ...	0 row(s) affected	0.032 sec
✓ 5	19:08:08	INSERT INTO Books (book_id, title, isbn, published_...	3 row(s) affected Records: 3 Duplicates: 0 Wamin...	0.015 sec
✓ 6	19:10:47	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.016 sec / 0.000 sec

Administration Schemas

No object selected

Step 5, 6 & 7 — Members Table: Create, Insert, Display

SCHEMAS

Filter objects

- bookflow_db
- bookflow_dbms
- college
- college_db
- college_dbms
- db
- dbms
- sys

Limit to 1000 rows

Jump to

```

1 • CREATE TABLE Members (
2   member_id INT PRIMARY KEY,
3   full_name VARCHAR(100),
4   email VARCHAR(100) UNIQUE
5 );
6

```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 1	19:06:49	drop database prac2	1 row(s) affected	0.032 sec
✓ 2	19:07:46	CREATE DATABASE prac2	1 row(s) affected	0.000 sec
✓ 3	19:07:46	USE prac2	0 row(s) affected	0.000 sec
✓ 4	19:07:51	CREATE TABLE Books (book_id INT PRIMARY ...	0 row(s) affected	0.032 sec
✓ 5	19:08:08	INSERT INTO Books (book_id, title, isbn, published_...	3 row(s) affected Records: 3 Duplicates: 0 Wamin...	0.015 sec
✓ 6	19:10:47	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.016 sec / 0.000 sec
✗ 7	19:12:17	CREATE TABLE Loans (loan_id INT PRIMARY K...	Error Code: 1824. Failed to open the referenced tabl...	0.000 sec
✓ 8	19:14:05	CREATE TABLE Members (member_id INT PRIM...	0 row(s) affected	0.406 sec

Administration Schemas

Filter objects

- bookflow_db
- bookflow_dbms
- college
- college_db
- college_dbms
- db
- dbms
- sys

```

1 • INSERT INTO Members (member_id, full_name, email) VALUES
2   (101, 'John Smith', 'john.smith@email.com'),
3   (102, 'Emma Wilson', 'emma.wilson@email.com'),
4   (103, 'Michael Brown', 'michael.brown@email.com');

```

Automatic context help is disabled. Use the toolbar to manually get help for the

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 1	19:06:49	drop database prac2	1 row(s) affected	0.032 sec
✓ 2	19:07:46	CREATE DATABASE prac2	1 row(s) affected	0.000 sec
✓ 3	19:07:46	USE prac2	0 row(s) affected	0.000 sec
✓ 4	19:07:51	CREATE TABLE Books (book_id INT PRIMARY ...	0 row(s) affected	0.032 sec
✓ 5	19:08:08	INSERT INTO Books (book_id, title, isbn, published...	3 row(s) affected Records: 3 Duplicates: 0 Wamin...	0.015 sec
✓ 6	19:10:47	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.016 sec / 0.000 sec
✗ 7	19:12:17	CREATE TABLE Loans (loan_id INT PRIMARY K...	Error Code: 1824. Failed to open the referenced tabl...	0.000 sec
✓ 8	19:14:05	CREATE TABLE Members (member_id INT PRIM...	0 row(s) affected	0.406 sec
✓ 9	19:15:18	INSERT INTO Members (member_id, full_name, em...	3 row(s) affected Records: 3 Duplicates: 0 Wamin...	0.062 sec

Filter objects

- bookflow_db
- bookflow_dbms
- college
- college_db
- college_dbms
- db
- dbms
- sys

```

1 • SELECT * FROM Members;

```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Result Grid

member_id	full_name	email
101	John Smith	john.smith@email.com
102	Emma Wilson	emma.wilson@email.com
103	Michael Brown	michael.brown@email.com
* NULL	NULL	NULL

Members 2 x Apply Revert

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 1	19:06:49	drop database prac2	1 row(s) affected	0.032 sec
✓ 2	19:07:46	CREATE DATABASE prac2	1 row(s) affected	0.000 sec
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✓ 4	19:07:51	CREATE TABLE Books (book_id INT PRIMARY ...	0 row(s) affected	0.032 sec
✓ 5	19:08:08	INSERT INTO Books (book_id, title, isbn, published...	3 row(s) affected Records: 3 Duplicates: 0 Wamin...	0.015 sec
✓ 6	19:10:47	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.016 sec / 0.000 sec
✗ 7	19:12:17	CREATE TABLE Loans (loan_id INT PRIMARY K...	Error Code: 1824. Failed to open the referenced tabl...	0.000 sec
✓ 8	19:14:05	CREATE TABLE Members (member_id INT PRIM...	0 row(s) affected	0.406 sec
✓ 9	19:15:18	INSERT INTO Members (member_id, full_name, em...	3 row(s) affected Records: 3 Duplicates: 0 Wamin...	0.062 sec
✓ 10	19:16:24	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

Step 8 — Create Loans Table (FOREIGN KEYS)

SCHEMAS

Filter objects

- bookflow_db
- bookflow_dbms
- college
- college_db
- college_dbms
- db
- dbms
- sys

Administration Schemas Information

No object selected

```

1 CREATE TABLE Loans (
2   loan_id INT PRIMARY KEY,
3   member_id INT,
4   book_id INT,
5   loan_date DATE,
6   FOREIGN KEY (member_id) REFERENCES Members(member_id),
7   FOREIGN KEY (book_id) REFERENCES Books(book_id)
8 );

```

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Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	19:06:49	drop database prac2	1 row(s) affected	0.032 sec
2	19:07:46	CREATE DATABASE prac2	1 row(s) affected	0.000 sec
3	19:07:46	USE prac2	0 row(s) affected	0.000 sec
4	19:07:51	CREATE TABLE Books (book_id INT PRIMARY ...	0 row(s) affected	0.032 sec
5	19:08:08	INSERT INTO Books (book_id, title, isbn, published...	3 row(s) affected Records: 3 Duplicates: 0 Wamin...	0.015 sec
6	19:10:47	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.016 sec / 0.000 sec
7	19:12:17	CREATE TABLE Loans (loan_id INT PRIMARY K...	Error Code: 1824. Failed to open the referenced tabl...	0.000 sec
8	19:14:05	CREATE TABLE Members (member_id INT PRIM...	0 row(s) affected	0.406 sec
9	19:15:18	INSERT INTO Members (member_id, full_name, em...	3 row(s) affected Records: 3 Duplicates: 0 Wamin...	0.062 sec
10	19:16:24	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
11	19:17:33	CREATE TABLE Loans (loan_id INT PRIMARY K...	0 row(s) affected	0.078 sec

Step 9 & 10 — Insert 10 Loans and Display

SCHEMAS

Filter objects

- bookflow_db
- bookflow_dbms
- college
- college_db
- college_dbms
- db
- dbms
- sys

Administration Schemas Information

No object selected

```

1 INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES
2   (1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'),
3   (3, 103, 3, '2025-01-10'), (4, 101, 2, '2025-02-01'),
4   (5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'),
5   (7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07'),
6   (9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');

```

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Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	19:06:49	drop database prac2	1 row(s) affected	0.032 sec
2	19:07:46	CREATE DATABASE prac2	1 row(s) affected	0.000 sec
3	19:07:46	USE prac2	0 row(s) affected	0.000 sec
4	19:07:51	CREATE TABLE Books (book_id INT PRIMARY ...	0 row(s) affected	0.032 sec
5	19:08:08	INSERT INTO Books (book_id, title, isbn, published...	3 row(s) affected Records: 3 Duplicates: 0 Wamin...	0.015 sec
6	19:10:47	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.016 sec / 0.000 sec
7	19:12:17	CREATE TABLE Loans (loan_id INT PRIMARY K...	Error Code: 1824. Failed to open the referenced tabl...	0.000 sec
8	19:14:05	CREATE TABLE Members (member_id INT PRIM...	0 row(s) affected	0.406 sec
9	19:15:18	INSERT INTO Members (member_id, full_name, em...	3 row(s) affected Records: 3 Duplicates: 0 Wamin...	0.062 sec
10	19:16:24	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
11	19:17:33	CREATE TABLE Loans (loan_id INT PRIMARY K...	0 row(s) affected	0.078 sec
12	19:21:39	INSERT INTO Loans (loan_id, member_id, book_id,...	10 row(s) affected Records: 10 Duplicates: 0 War...	0.016 sec

Step 11 — JOIN Query

SCHEMAS

Filter objects

bookflow_db

bookflow_dbms

college

college_db

college_dbms

db

dbms

sys

Administration

Schemas

Information

No object selected

Limit to 1000 rows

Jump to

1 • START TRANSACTION;

2

3 • INSERT INTO Books (book_id, title, isbn, published_year)

4 VALUES (4, 'Animal Farm', '9780451526342', 1945);

5

6 • INSERT INTO Donation_History (donation_id, book_id, donor_name,

7 VALUES (1, 4, 'Raj Kumar', CURDATE());

8

9 • COMMIT;

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
4	19:07:51	CREATE TABLE Books (book_id INT PRIMARY...	0 row(s) affected	0.032 sec
5	19:08:08	INSERT INTO Books (book_id, title, isbn, publish...	3 row(s) affected Records: 3 Duplicates: 0 Warni...	0.015 sec
6	19:10:47	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.016 sec / 0.000 sec
7	19:12:17	CREATE TABLE Loans (loan_id INT PRIMARY ...	Error Code: 1824. Failed to open the referenced ta...	0.000 sec
8	19:14:05	CREATE TABLE Members (member_id INT PRI...	0 row(s) affected	0.406 sec
9	19:15:18	INSERT INTO Members (member_id, full_name, e...	3 row(s) affected Records: 3 Duplicates: 0 Warni...	0.062 sec
10	19:16:24	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
11	19:17:33	CREATE TABLE Loans (loan_id INT PRIMARY ...	0 row(s) affected	0.078 sec
12	19:21:39	INSERT INTO Loans (loan_id, member_id, book_j...	10 row(s) affected Records: 10 Duplicates: 0 Wa...	0.016 sec
13	19:23:03	SELECT m.full_name AS Member_Name, b.title A...	10 row(s) returned	0.000 sec / 0.000 sec
14	19:24:40	SELECT published_year, COUNT(book_id) AS To...	3 row(s) returned	0.000 sec / 0.000 sec
15	19:26:08	CREATE TABLE Donation_History (donation_id ...	0 row(s) affected	0.062 sec
16	19:26:08	CREATE TABLE Donation_History (donation_id ...	Error Code: 1050. Table 'donation_history' already ...	0.000 sec
17	19:26:26	CREATE TABLE Donation_History (donation_id ...	Error Code: 1050. Table 'donation_history' already ...	0.000 sec
18	19:29:19	START TRANSACTION	0 row(s) affected	0.000 sec
19	19:29:19	INSERT INTO Books (book_id, title, isbn, publish...	1 row(s) affected	0.000 sec
20	19:29:19	INSERT INTO Donation_History (donation_id, bo...	1 row(s) affected	0.000 sec
21	19:29:19	COMMIT	0 row(s) affected	0.000 sec

M.Keerthi Sri Local instance (...)

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

SQL File 3*

```

1 CREATE TABLE Donation_History (
2     donation_id INT PRIMARY KEY,
3     book_id INT,
4     donor_name VARCHAR(100),
5     donation_date DATE,
6     FOREIGN KEY (book_id) REFERENCES Books(book_id)
7 );

```

Output

Action Output

#	Time	Action	Message
1	12:26:02	DROP DATABASE bookflow_db	0 row(s) affected
2	12:26:10	CREATE DATABASE bookflow_db	1 row(s) affected
3	12:26:10	USE bookflow_db	0 row(s) affected
4	12:29:05	CREATE TABLE Books (book_id INT PRIMARY...	0 row(s) affected
5	12:29:42	INSERT INTO Books (book_id, title, isbn, published...	3 row(s) affected Records: 3
6	12:30:59	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned
7	12:32:46	INSERT INTO Members (member_id, full_name, em...	Error Code: 1146. Table 'book...
8	12:33:05	INSERT INTO Members (member_id, full_name, em...	Error Code: 1146. Table 'book...
9	12:33:37	CREATE TABLE Members (member_id INT PRI...	0 row(s) affected
10	12:34:02	INSERT INTO Members (member_id, full_name, em...	3 row(s) affected Records: 3
11	12:34:24	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned
12	12:35:05	CREATE TABLE Loans (loan_id INT PRIMARY ...	0 row(s) affected
13	12:39:54	INSERT INTO Loans (loan_id, member_id, book_id...	Error Code: 1264. Out of range
14	13:27:13	INSERT INTO Loans (loan_id, member_id, book_id...	Error Code: 1452. Cannot add
15	13:29:38	INSERT INTO Loans (loan_id, member_id, book_id...	10 row(s) affected Records: 10
16	13:29:45	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned
17	13:30:33	SELECT m.full_name AS Member_Name, b.titl...	10 row(s) returned
18	13:31:02	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned
19	13:31:40	SELECT m.full_name AS Member_Name, b.titl...	10 row(s) returned
20	13:32:03	SELECT published_year, COUNT(book_id) AS...	3 row(s) returned
21	13:32:25	CREATE TABLE Donation_History (donation_id ...	0 row(s) affected

Step 17 & 18 — Index on ISBN + Fast Search

Filter objects

- bookflow_db
- bookflow_dbms
- college
- college_db
- college_dbms
- db
- dbms
- sys

Administration Schemas

Information

```

1 CREATE INDEX idx_books_isbn ON Books(isbn);
2 SELECT * FROM Books WHERE isbn = '9780451524935';
3 drop database prac2;

```

Result Grid

book_id	title	isbn	published_year
3	1984	9780451524935	1949
NULL	NULL	NULL	NULL

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid

Form Editor

Field Types

Query Stats